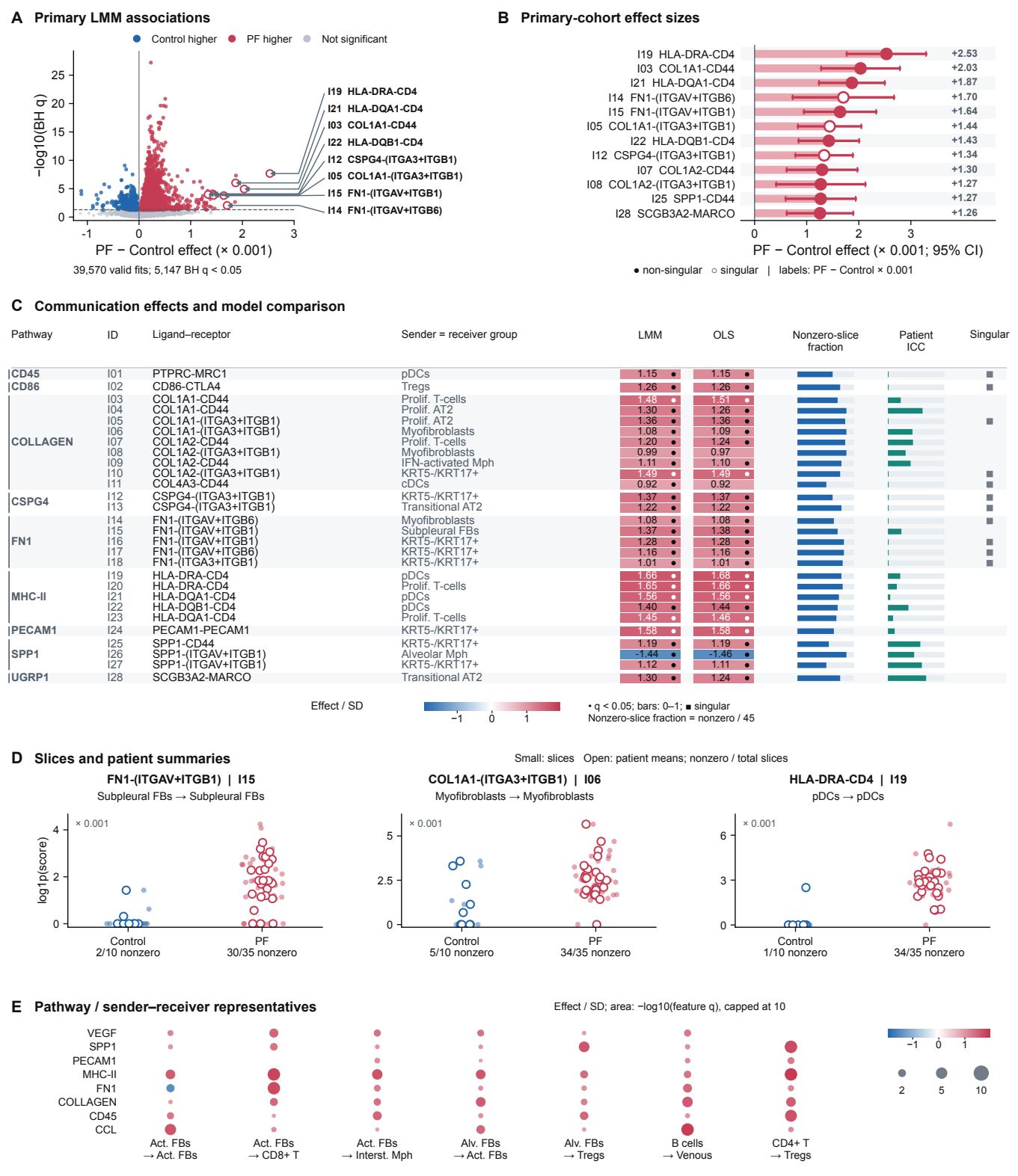

A Primary LMM associations
Control higher
PF higher
Not significant
−log10(BH q)
PF − Control effect (× 0.001)
39,570 valid fits; 5,147 BH q < 0.05
I19 HLA-DRA-CD4
I21 HLA-DQA1-CD4
I03 COL1A1-CD44
I22 HLA-DQB1-CD4
I12 CSPG4-(ITGA3+ITGB1)
I05 COL1A1-(ITGA3+ITGB1)
I15 FN1-(ITGAV+ITGB1)
I14 FN1-(ITGAV+ITGB6)
B Primary-cohort effect sizes
I19 HLA-DRA-CD4 +2.53
I03 COL1A1-CD44 +2.03
I21 HLA-DQA1-CD4 +1.87
I14 FN1-(ITGAV+ITGB6) +1.70
I15 FN1-(ITGAV+ITGB1) +1.64
I05 COL1A1-(ITGA3+ITGB1) +1.44
I22 HLA-DQB1-CD4 +1.43
I12 CSPG4-(ITGA3+ITGB1) +1.34
I07 COL1A2-CD44 +1.30
I08 COL1A2-(ITGA3+ITGB1) +1.27
I25 SPP1-CD44 +1.27
I28 SCGB3A2-MARCO +1.26
PF − Control effect (× 0.001; 95% CI)
● non-singular ○ singular | labels: PF − Control × 0.001
C Communication effects and model comparison
Pathway
ID
Ligand–receptor
Sender = receiver group
LMM
OLS
Nonzero-slice fraction
Patient ICC
Singular
CD45 I01 PTPRC-MRC1 pDCs 1.15 1.15
CD86 I02 CD86-CTLA4 Tregs 1.26 1.26
COLLAGEN I03 COL1A1-CD44 Prolif. T-cells 1.48 1.51
I04 COL1A1-CD44 Prolif. AT2 1.30 1.26
I05 COL1A1-(ITGA3+ITGB1) Prolif. AT2 1.36 1.36
I06 COL1A1-(ITGA3+ITGB1) Myofibroblasts 1.08 1.09
I07 COL1A2-CD44 Prolif. T-cells 1.20 1.24
I08 COL1A2-(ITGA3+ITGB1) Myofibroblasts 0.99 0.97
I09 COL1A2-CD44 IFN-activated Mph 1.11 1.10
I10 COL1A2-(ITGA3+ITGB1) KRT5-/KRT17+ 1.49 1.49
I11 COL4A3-CD44 cDCs 0.92 0.92
CSPG4 I12 CSPG4-(ITGA3+ITGB1) KRT5-/KRT17+ 1.37 1.37
I13 CSPG4-(ITGA3+ITGB1) Transitional AT2 1.22 1.22
FN1 I14 FN1-(ITGAV+ITGB6) Myofibroblasts 1.08 1.08
I15 FN1-(ITGAV+ITGB1) Subpleural FBs 1.37 1.38
I16 FN1-(ITGAV+ITGB1) KRT5-/KRT17+ 1.28 1.28
I17 FN1-(ITGAV+ITGB6) KRT5-/KRT17+ 1.16 1.16
I18 FN1-(ITGA3+ITGB1) KRT5-/KRT17+ 1.01 1.01
MHC-II I19 HLA-DRA-CD4 pDCs 1.66 1.68
I20 HLA-DRA-CD4 Prolif. T-cells 1.65 1.66
I21 HLA-DQA1-CD4 pDCs 1.56 1.56
I22 HLA-DQB1-CD4 pDCs 1.40 1.44
I23 HLA-DQA1-CD4 Prolif. T-cells 1.45 1.46
PECAM1 I24 PECAM1-PECAM1 KRT5-/KRT17+ 1.58 1.58
SPP1 I25 SPP1-CD44 KRT5-/KRT17+ 1.19 1.19
I26 SPP1-(ITGAV+ITGB1) Alveolar Mph -1.44 -1.46
I27 SPP1-(ITGAV+ITGB1) KRT5-/KRT17+ 1.12 1.11
UGRP1 I28 SCGB3A2-MARCO Transitional AT2 1.30 1.24
Effect / SD
• q < 0.05; bars: 0–1; ■ singular
Nonzero-slice fraction = nonzero / 45
D Slices and patient summaries
Small: slices Open: patient means; nonzero / total slices
FN1-(ITGAV+ITGB1) | I15
Subpleural FBs → Subpleural FBs
log1p(score)
Control 2/10 nonzero
PF 30/35 nonzero
COL1A1-(ITGA3+ITGB1) | I06
Myofibroblasts → Myofibroblasts
Control 5/10 nonzero
PF 34/35 nonzero
HLA-DRA-CD4 | I19
pDCs → pDCs
Control 1/10 nonzero
PF 34/35 nonzero
E Pathway / sender–receiver representatives
Effect / SD; area: −log10(feature q), capped at 10
VEGF
SPP1
PECAM1
MHC-II
FN1
COLLAGEN
CD45
CCL
Act. FBs → Act. FBs
Act. FBs → CD8+ T
Act. FBs → Interst. Mph
Alv. FBs → Act. FBs
Alv. FBs → Tregs
B cells → Venous
CD4+ T → Tregs